



Data Integration Platform Characteristic Trait

Abstract:

In recent times, for every business, the associated data volume has been quite large and complex in nature. The sources that generate these data are quite important and need to be understood by businesses, and according to the data pattern, businesses need to realign or take important decisions. However, the data types originated from diverse sources are not similar; hence, every organization needs its data handling capacity to grow up to use the heterogeneous data source by implementing a modern data integration tool that is able to optimize the data management capabilities and strategies.

The data generated from scattered sources is heterogeneous in nature and holds different formats of data; hence, it is quite complex to apply business rules and perform data analysis.

Hence, choosing the right data integration tool to keep up with the changing pace and apply automation features is quite important to driving the data integration space. These features should be able to reduce the overall project development time and easily outline the complex business areas. Data integration tools are capable of moving sets of data from various sources; they sometimes apply themselves to a few business rules and other times they don't change, but the metadata around the data changes from source to target system. Meeting with these norms is quite complex and tedious some time where integration tool came to a savior.

Authors

Bhabani Prasad Upadhyaya

(Senior Architect-Practice-GP-App Services & Cloud)

Sasken Technologies Ltd.



Table of Content

Introduction: Data Integration platform and their capabilities like ETL/ELT, Data Replication & Change Data Capture	4
Integration tool platform in the space of Self-service	6
Collaboration in the Data Integration Platform viz mainly in the self-service area	7
Automation in Data Integration platform	8
Innovation and Acceleration in Data Integration platform	9
Forrester wave capability studies on the Data Integration Tool	10
Conclusion	11
Reference	12
About the Author	12
About Sasken	12



Introduction: Data Integration platform and their capabilities like ETL/ELT, Data Replication & Change Data Capture

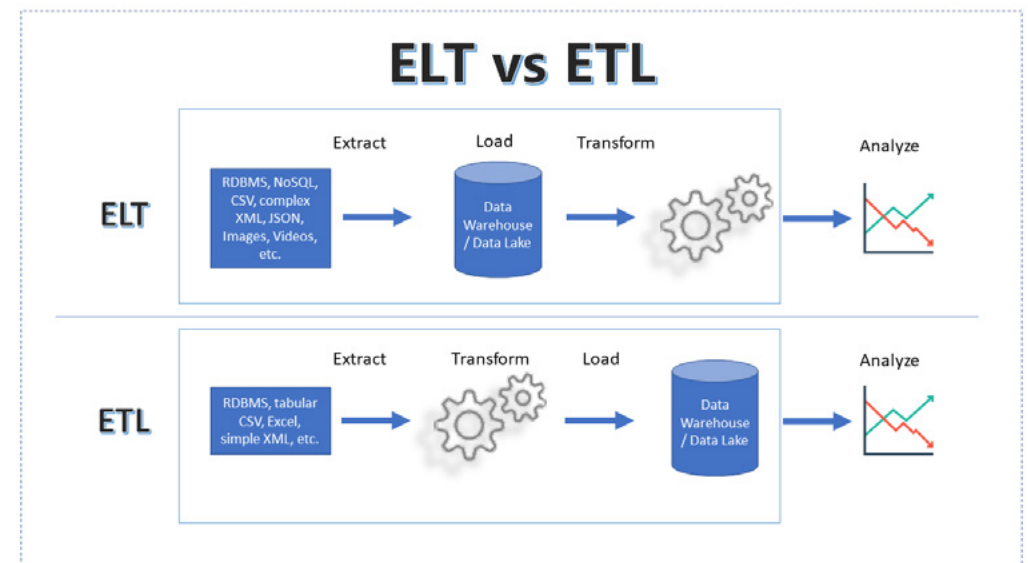
Data is quite important in any business; it is required to be analyzed and understood by the business to take corrective measures for business decisions. Hence, data needs to be built with a lot of business transformation rules applied on top of it while also maintaining the requirements of the system where the data resides. Moving the data from the source to the target system sounds easy; however, the movement sometimes involves complex transformations, like the cleansing of the data to derive meaningful data from the source, applying transformations, and applying the data on the target with an incremental basis of load, i.e., applying the data in a CDC (Change Data Capture) perspective.

Sometimes the data remains the same from source to target, but the metadata around different systems changes, so it is necessary to copy and move the source system data to the target system while transforming the metadata to meet the target system definitions.

Because the fundamental rule or action for an integration tool is to satisfy the ETL capabilities (Extract, Transform, and Load), most of the time, the data integration tool itself is called an ETL tool, or to a lesser extent, an ELT tool, where it stands for Extraction, Load, and Transformation. ELT differs from ETL in the process of transformation being applied to it. Hence, it is necessary to thoroughly examine

which process should be followed for data integration, depending on paradigms like architectural, performance, and security insinuations.

Hence, data integration is the core of the service to satisfy a robust and easiest ETL/ELT platform capable of moving data across systems.



The metadata surrounding your data will often need to change as the data itself remains the same.

(Source: Bloor Research International Ltd.)

Addressing other integration practices like replication, CDC, etc. followed in a data warehouse scenario.

In fewer cases, the data needs to be copied from source to target without many changes, like propagating the change data; these can be achieved via different technologies like

- CDC: Change Data Capture is used to monitor the source data for any changed information and propagate the change to the target system. Doing this method of transferring data keeps the downstream system updated with respect to the source, though it's not streaming data. However, it is near real time. CDC can integrate with an ETL or ELT process to apply transformation to meet the business aspects of data.
- Data replication is another mechanism where a huge volume of data needs to move from source to target, like moving traditional databases from premises to the cloud. In this kind of scenario, the data movement happens once to keep the target with the most recent data, so replication and the CDC mechanism hold up well.

- Data integration changed over time, like adaptation to cloud-specific technology, which evolved to support the recent trend and cycle seen for cloud-specific migration or tools. Data integration tools make the target system available with meaningful data that can be pulled via the reporting tool for data visualization.
- With changing trends and business needs, now multiple enterprises need the data before it runs for a transformation. This is where data virtualization comes in; instead of moving the data from different subsystems, business users can connect and run their analytical queries to obtain the data they need.

Data virtualization became more important and was included in the data integration tool ecosystem, where before transformation it can be plugged out or run with AI/ML-related capabilities to build a model that is then used for pattern analysis.

Data virtualization provides a viable alternative to data integration by allowing you to query data that exists in multiple physical locations.

(Source: Bloor Research International Ltd.)



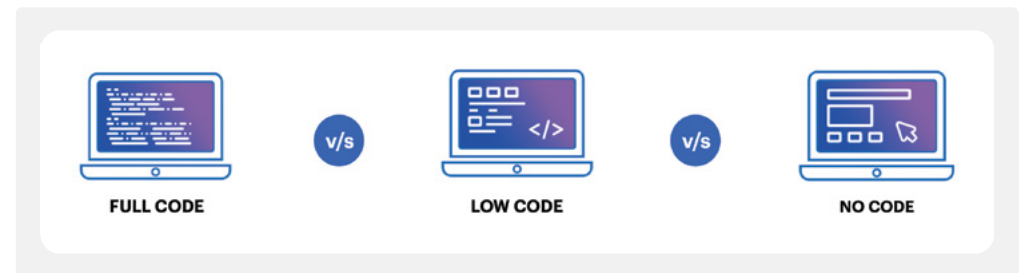
Integration tool platform in the space of Self-service

Self-service data integration is a process where business users can connect various data sources, transform and manipulate the data, and publish the integrated data sets without much help from the IT team. This gives an edge to making data-driven decisions in real-time.

This needs to be a simple, usable visual interface where the user can drag-and-drop functionality to allow for the creation of data models, data pipelines, and automated pipeline-related workflows. A self-service integration tool also simplifies the data integration process by identifying and mapping the data elements across multiple systems.

The self-service aspect of data integration is that it allows the data consumer to move the data as per their needs. They can build data models and retrieve useful business information without involving IT, allowing IT and businesses to work more efficiently.

The self-service data integration tool uses a visual interface and drag-and-drop functionality to allow non-technical users to create a data map, transform the data, and automate the workflow. It offers a low-code to no-code platform where the business user uses the data functionality for their needs. These functions' features are rapidly evolving and are likely to be widely adopted in the data sphere of work.



There is an increasing trend toward provision of the self-service capabilities within data integration space.

(Source: Bloor Research International Ltd.)



Collaboration in the Data Integration Platform viz mainly in the self-service area

Collaboration self-service is a powerful tool in the self-service domain where multiple business users work together on data sharing and workflows. Users are also able to leave comments on any given data asset. This helps bring domain expertise, which resides within the organization, in a centralized and systemic way. This is mainly useful to retain the functional knowledge around the data pipeline. Data pipelines are data assets that automate the data flow, but collaboration can bring domain value to the data pipelines.



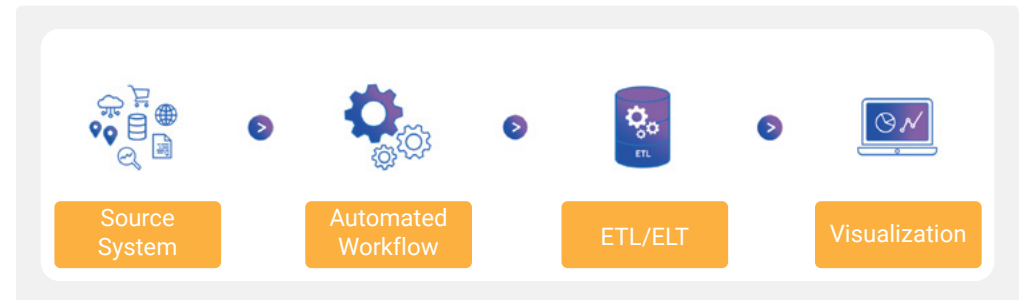
Collaboration serves as a natural step forward from self-service.
(Source: Bloor Research International Ltd.)



Automation in Data Integration platform

Automation is the key for any industry to optimize cost and project execution with ease. It always remains a prime concern for data integration, like in a big data warehouse where you run a reconciliation report every day to find out the missing data and chase up with the source team that missed the feed on a prior day. It's quite challenging and quite tough to implement the logic.

Automation is quite important in any development project activity, so as in the data integration platform, the automation capabilities in a data integration tool or product can be a clear differentiator amongst others. The automation features can be anything as per the customer's or business's needs; however, as a standard offering, they can incorporate machine learning throughout their built-in data processes and provide extensible automation capabilities with other platforms or tools.



A highly automated solution will feature a variety of embedded automation and machine learning throughout its built-in data processes.
(Source: Bloor Research International Ltd.)



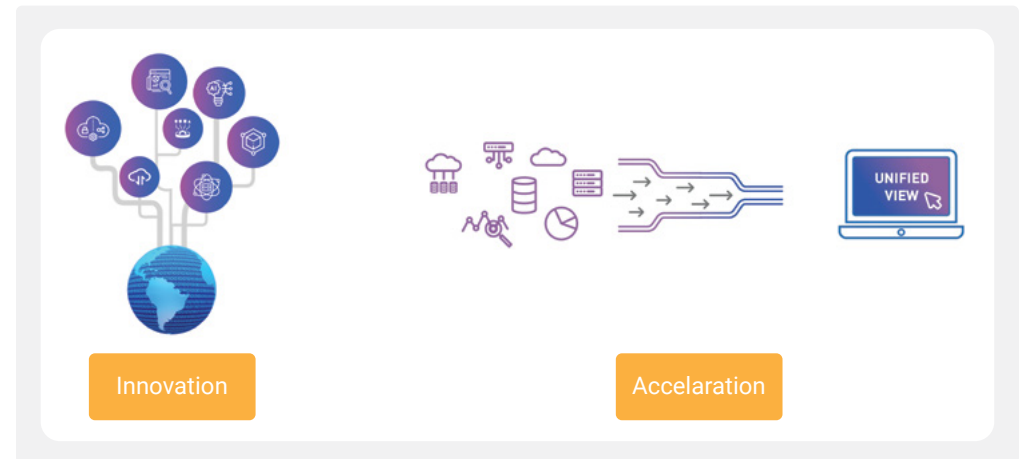
Innovation and Acceleration in Data Integration platform

Data integration is a mature space. Having said this, it is closer to being stagnant. However, the data generated over the years has become more volatile and complex, so innovation is necessary to take advantage of modern data trends and maximize the benefits for business users even in a mature space like data integration.

As a result, proving a willingness to innovate is quite critical and crucial for a data integration platform.

Combining all above critical qualities and putting inside to a data platform tool bring flexibility to define the data lineage i.e., starting from data extraction to transformation, also visualize the data by different stake holders which bring them data insightful and place them ahead understanding the business demand.

Automation of data Pipelines, extending the self-service solution, i.e., the low-code to no-code platform, and placing it under a single umbrella as a data integration tool, increase overall speed and efficiency where you can move data within the system, which enables users to extract value insight from data more quickly, more readily, and at a greater volume.



Data integration will continue to be an essential part of businesses innovation, and this will help business improve the whole experience for the consumer.

Combining everything like automation, low-code to no-code, etc., will dramatically increase the overall speed and efficiency at which businesses can move and transform data.

(Source: Bloor Research International Ltd.)



Forrester wave capability studies on the Data Integration Tool



Source: Gartner (August 2022)



Conclusion

Data integration platforms are a mature space where more than one integration platform has laid the functionalities to cater for the data integration solution.

This document enables you to figure out the right tool for your data integration space with respect to the modern-day data landscape.

To outline the most desirable characteristic of the data integration platform, it should have the below-highlighted functionality, not limited to only the below, but as a must-have to match modern data trends.

Serial No	Functionality	Features must have
1.	Supporting AI/ML based Platforms.	YES
2.	Having more of in-built connectors to connect different data source.	YES
3.	Low-code to No-code platform – Self-service based platform.	YES
4.	Troubleshoot with clicks.	YES
5.	Modern cloud native architecture.	YES
6.	Capability to have batch and real time data integration.	YES
7.	Built in Dev-Ops, Data-Ops and ML-Ops capability.	YES

8.	Full Automated data pipelines.	YES
9.	Enterprise based support on stability, reliability, and security.	YES
10.	Automated workflow and Pipeline Orchestration.	YES
11.	Open and Cloud agnostic interoperability possibility.	YES
12.	Built in Template for Data Ingestion for Batch and near real time like Change data capture.	YES



Reference

- Bloor Research International Limited
- Gartner Inc
- Gathr One
- Various Data Warehouse Inference

About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation, providing concept-to-market and chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Consumer Electronics, Enterprise Devices, SatCom, Telecom, and Transportation.

About the Author

Bhabani Prasad Upadhyaya has more than 22 years of experience in ITES-enabled sectors, working in different industry verticals and delivering high-quality, high-throughput data-driven solutions. Here at Sasken, he works with data practices, GP-App Services, and cloud-specific domains.





Data Integration Platform Characteristic Trait

marketing@sasken.com | www.sasken.com

USA | FINLAND | GERMANY | JAPAN | INDIA

© Sasken Technologies Ltd. All rights reserved.

Products and services mentioned herein are trademarks and service marks of **Sasken Technologies Ltd.**, or the respective companies.